

Proof of Lemma 0.1.3.

Let $P^* \geq P$ be a refinement of P . Since partition contains only finite elements,

WLOG, assume that $P^* = P \cup \{S\}$ s.t. $S \notin P = \{x_0, \dots, x_n\}$

Then, for some $1 \leq j \leq n$ $x_{j-1} < S < x_j$. Now,

$$\begin{aligned} U(f, P, \alpha) - U(f, P^*, \alpha) &= \sup_{t \in [x_{j-1}, x_j]} f(t) \cdot [\alpha(x_j) - \alpha(S)] + \sup_{t \in [x_{j-1}, x_j]} f(t) \cdot [\alpha(S) - \alpha(x_{j-1})] \\ &\quad - \sup_{t \in [x_{j-1}, S]} f(t) \cdot [\alpha(x_j) - \alpha(S)] - \sup_{t \in [S, x_j]} f(t) \cdot [\alpha(S) - \alpha(x_{j-1})] \\ &\geq 0. \quad \square \end{aligned}$$

Proof of Lemma 0.1.4. X fail try.

Since $f, g \in R(\alpha)$, $\int_a^b f \, d\alpha = \int_a^b f \, d\alpha$ and $\int_a^b g \, d\alpha = \int_a^b g \, d\alpha$. Now,

$$\begin{aligned} \int_a^b f + g \, d\alpha &= \inf_{P \in \mathcal{P}_{[a,b]}} U(f+g, P, \alpha) = \inf_{P \in \mathcal{P}_{[a,b]}} \left[\sum_{i=1}^n M_i(f+g) \cdot \Delta\alpha_i \right] \\ &\leq \inf_{P \in \mathcal{P}_{[a,b]}} \left[\sum_{i=1}^n M_i(f) \cdot \Delta\alpha_i + \sum_{i=1}^n M_i(g) \cdot \Delta\alpha_i \right] \end{aligned}$$

Proof of Thm 0.1.4

1) $\Rightarrow$ 2). Since $f \in \mathcal{R}(a)$, given $\varepsilon > 0$, there exists a partition $P \in \mathcal{P}[a, b]$ s.t.

$$\int_a^b f \, d\alpha \leq U(f, P, \alpha) < \int_a^b f \, d\alpha + \frac{\varepsilon}{2} \quad \text{and} \quad \int_a^b f \, d\alpha \geq L(f, P, \alpha) > \int_a^b f \, d\alpha - \frac{\varepsilon}{2}$$

Ask the next step to a toddler on the street.

2) $\Rightarrow$ 1). It is clear, because for any partition $P \in \mathcal{P}[a, b]$,

$$\int_a^b f \, d\alpha - \int_a^b f \, d\alpha \leq U(f, P, \alpha) - L(f, P, \alpha)$$

To show 1) $\Leftrightarrow$ 3), we prove a statement: If 2) holds, then for any $1 \leq i \leq n$,

$$\text{for arbitrary } s_i, t_i \in [x_{i-1}, x_i], \quad \sum_{i=1}^n |f(s_i) - f(t_i)| \cdot \Delta\alpha_i < \varepsilon.$$

$$\text{Since } |f(s_i) - f(t_i)| \leq M_i - m_i,$$

$$\sum_{i=1}^n |f(s_i) - f(t_i)| \cdot \Delta\alpha_i \leq U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon.$$

1) $\Rightarrow$ 3). Suppose $f \in \mathcal{R}(a)$. Then, by 2), there exists a partition $P_0 \in \mathcal{P}[a, b]$ s.t.

$$U(f, P_0, \alpha) - L(f, P_0, \alpha) < \varepsilon$$

Now, for any refinement $P > P_0$,

$$U(f, P, \alpha) - L(f, P, \alpha) \leq U(f, P_0, \alpha) - L(f, P_0, \alpha) < \varepsilon$$

Clearly, regardless of choice of t_i ,

$$L(f, P, \alpha) \leq S(f, P, \alpha) \leq U(f, P, \alpha)$$

$$\text{and} \quad U(f, P, \alpha) \geq \int f \, d\alpha > L(f, P, \alpha)$$

$$\text{Therefore, } \left| S(f, P, \alpha) - \int f \, d\alpha \right| \leq U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon.$$

3) $\Rightarrow$ 1) is trivial.

Proof of lemma 0.1.5.

Let $f, g \in \mathcal{R}(a)$ be given. Given $\varepsilon > 0$, there exists a partition $P \in \mathcal{P}[a, b]$ such that

$$U(f, P, \alpha) - L(f, P, \alpha) < \frac{\varepsilon}{2} \quad \text{and} \quad U(g, P, \alpha) - L(g, P, \alpha) < \frac{\varepsilon}{2}$$

(Technically, take $P_1, P_2 \in \mathcal{P}[a, b]$ for f, g respectively, and put $P = P_1 \cup P_2$). Now,

$$U(f+g, P, \alpha) - L(f+g, P, \alpha) \leq U(f, P, \alpha) + U(g, P, \alpha) - L(f, P, \alpha) - L(g, P, \alpha) < \varepsilon.$$

And the equality is obtained by: Given $\varepsilon > 0$, there exists a partition $P \in \mathcal{P}[a, b]$ s.t.

$$\begin{aligned} \int_a^b f \, d\alpha + \int_a^b g \, d\alpha - \varepsilon &\leq L(f, P, \alpha) + L(g, P, \alpha) \leq L(f+g, P, \alpha) \\ &\leq \int_a^b f+g \, d\alpha \\ &\leq U(f+g, P, \alpha) \leq U(f, P, \alpha) + U(g, P, \alpha) \leq \int_a^b f \, d\alpha + \int_a^b g \, d\alpha + \varepsilon \end{aligned}$$

The second assertion is clear. Now, we can say that

$\mathcal{R}(a)$ is a $\mathbb{R}$ -vector space.

Proof of lemma 0.1.6.

Suppose that $f \in \mathcal{R}(a)$ on $[a, b]$. Given $\varepsilon > 0$, there exists $P \in \mathcal{P}[a, b]$ s.t.

$$U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon.$$

Put $P^* = P \cup \{c\}$. Then, for $P_1 = P^* \cap [a, c]$ and $P_2 = P^* \cap [c, b]$. Now,

$$\begin{aligned} \int_a^c f \, d\alpha - \int_a^c f \, d\alpha + \int_c^b f \, d\alpha - \int_c^b f \, d\alpha &\leq U(f, P_1, \alpha) - L(f, P_1, \alpha) + U(f, P_2, \alpha) - L(f, P_2, \alpha) \\ &= U(f, P^*, \alpha) - L(f, P^*, \alpha) < \varepsilon \end{aligned}$$

Similarly, the converse holds.

Proof of Thm 0.1.8.

Given $\epsilon > 0$, take $\eta > 0$ s.t. $[\alpha(b) - \alpha(a)] \cdot \eta < \epsilon$.

Since f is continuous on the compact set $[a, b]$, it is continuous uniformly.

Thus, there exists $\delta > 0$ s.t.

$$\forall x, y \in [a, b], |x - y| < \delta \Rightarrow |f(x) - f(y)| < \eta.$$

Now, let $P \in \mathcal{P}[a, b]$ be a partition such that for all i , $\Delta x_i < \delta$ ($P = \{x_0, \dots, x_n\}$)

Then, for all i , $M_i - m_i \leq \eta$. Now,

$$U(f, P, \alpha) - L(f, P, \alpha) = \sum_{i=1}^n (M_i - m_i) \cdot \Delta \alpha_i \leq \sum_{i=1}^n \eta \cdot \Delta \alpha_i = \eta \cdot \sum_{i=1}^n \Delta \alpha_i = \eta \cdot [\alpha(b) - \alpha(a)] < \epsilon.$$

only

Proof of Thm 0.1.9. Alternatively, since monotone function can have ^{only} countably simple discontinuity, if no such shared discontinuities with f and α , then countably discontinuous f is R.S. integrable. But, in this, prove by elementary, using properties of monotone.

Since α is continuous, for each $n \in \mathbb{N}$, there exists a partition $P_n \in \mathcal{P}[a, b]$,

$$\Delta \alpha_i = \frac{\alpha(b) - \alpha(a)}{n} (= \Delta \alpha) \text{ for all } i$$

(More specifically, let $n \in \mathbb{N}$ be fixed. By the Intermediate Value Theorem, there exists $x_1 \in [a, b]$ s.t. $\alpha(x_1) = \alpha(a) + \frac{\alpha(b) - \alpha(a)}{n}$. Inductively,

there exists $x_2 \in [x_1, b]$ s.t. $\alpha(x_2) = \alpha(a) + 2 \cdot \frac{\alpha(b) - \alpha(a)}{n}$

$\vdots$

$x_{n-1} \in [x_{n-2}, b]$ s.t. $\alpha(x_{n-1}) = \alpha(a) + (n-1) \cdot \frac{\alpha(b) - \alpha(a)}{n}$

And take $x_n = b$. Essentially, the existence such partition is obtained by I.V.P.

But, Monotone can have only first-kind discontinuity and

I.V.P. implies it can not have first-kind discontinuity, thus Monotone I.V.P. $\Rightarrow$ Continuity.

wlog,

Given $\epsilon > 0$, Assume f is monotone increasing. Take $n \in \mathbb{N}$ s.t. $\frac{[\alpha(b) - \alpha(a)] \cdot [f(b) - f(a)]}{n} < \epsilon$.

$$U(f, P_n, \alpha) - L(f, P_n, \alpha) = \sum_{i=1}^n [M_i - m_i] \cdot \Delta \alpha_i = \sum_{i=1}^n [f(x_i) - f(x_{i-1})] \Delta \alpha_i = \Delta \alpha \cdot \sum_{i=1}^n [f(x_i) - f(x_{i-1})]$$

$$= \Delta \alpha \cdot [f(b) - f(a)] < \epsilon. \quad \square$$

Proof of Thm. 0.1.10.

Since ϕ is continuous on the compact set $[m, M]$, it is continuous uniformly.

Given $\epsilon > 0$, there exists $0 < \delta < \epsilon$ s.t. $|s - t| < \delta \Rightarrow |\phi(s) - \phi(t)| < \epsilon$.

Since $f \in \mathcal{R}(\alpha)$ on $[a, b]$, there is a partition $\{x_0, \dots, x_n\}$ on $[a, b]$ s.t.

$$U(f, P, \alpha) - L(f, P, \alpha) < \delta^2$$

Let $M_i = \sup f(t)$ and $m_i = \inf f(t)$ and $M_i^* = \sup(\phi \circ f(t))$ and $m_i^* = \inf(\phi \circ f(t))$ where the sup, inf values are taken in $t \in [x_{i-1}, x_i]$.

Divides the index set $\{1, \dots, n\}$ as:

$$A = \{i \in \{1, \dots, n\} \mid M_i - m_i < \delta\} \text{ and } B = \{i \in \{1, \dots, n\} \mid M_i - m_i \geq \delta\}.$$

Now, if $i \in A$, then $M_i^* - m_i^* = \sup(\phi \circ f(t)) - \inf(\phi \circ f(t)) \leq \epsilon$

because $\forall s, t \in [x_{i-1}, x_i], |f(s) - f(t)| \leq M_i - m_i \Rightarrow |\phi(f(s)) - \phi(f(t))| < \epsilon$.

And, if $i \in B$, then $M_i^* - m_i^* \leq 2 \cdot \sup_{t \in [a, b]} |\phi(t)|$

We obtain that

$$\delta \cdot \sum_{i \in B} \Delta \alpha_i \leq \sum_{i \in B} [M_i - m_i] \cdot \Delta \alpha_i \leq \sum_{i=1}^n [M_i - m_i] \cdot \Delta \alpha_i = U(f, P, \alpha) - L(f, P, \alpha) < \delta^2.$$

That is $\sum_{i \in B} \Delta \alpha_i < \delta$. Finally,

$$\begin{aligned} \sum_{i=1}^n [M_i^* - m_i^*] \Delta \alpha_i &= \sum_{i \in A} [M_i^* - m_i^*] \Delta \alpha_i + \sum_{i \in B} [M_i^* - m_i^*] \Delta \alpha_i \\ &\leq \epsilon \cdot \sum_{i \in A} \Delta \alpha_i + 2 \cdot \sup_{t \in [m, M]} |\phi(t)| \cdot \sum_{i \in B} \Delta \alpha_i \\ &\leq \epsilon \cdot [\alpha(b) - \alpha(a)] + 2 \sup |\phi| \cdot \delta \\ &\leq \epsilon \cdot [\alpha(b) - \alpha(a) + 2 \sup |\phi|] \end{aligned}$$

Proof of Thm 8.1.11.

Suppose that $f \in \mathcal{R}(a)$. Given $\epsilon > 0$, there is a partition $P = \{x_0, \dots, x_n\}$ on $[a, b]$ s.t.
 $U(f, P) - L(f, P) < \epsilon$, since f is continuous, thus $f' \in \mathcal{R}$.

By the Mean-Value Theorem, for each $1 \leq i \leq n$

$$\Delta \alpha_i = \alpha(x_i) - \alpha(x_{i-1}) = \alpha'(t_i)(x_i - x_{i-1}) = \alpha'(t_i) \cdot \Delta x_i \text{ for some } t_i \in [x_{i-1}, x_i]$$

For any $s_i \in [x_{i-1}, x_i]$,

$$\sum_{i=1}^n |\alpha'(s_i) - \alpha'(t_i)| \cdot \Delta x_i \leq \sum_{i=1}^n [M_i - m_i] \cdot \Delta x_i < \epsilon.$$

$$\begin{aligned} \text{Now, } & \left| \sum_{i=1}^n f(s_i) \cdot \Delta \alpha_i - \sum_{i=1}^n f(s_i) \cdot \alpha'(s_i) \cdot \Delta x_i \right| \\ &= \left| \sum_{i=1}^n f(s_i) \cdot \alpha'(t_i) \cdot \Delta x_i - \sum_{i=1}^n f(s_i) \alpha'(s_i) \cdot \Delta x_i \right| \\ &= \left| \sum_{i=1}^n [\alpha'(t_i) - \alpha'(s_i)] \cdot f(s_i) \cdot \Delta x_i \right| \leq \sup |f| \cdot \epsilon \quad (*) \end{aligned}$$

$$\text{Therefore, } \left| \int_a^b f \, d\alpha - \int_a^b f \alpha' \, dx \right| \leq |U(f, P, \alpha) - U(f \alpha', P, x)| \leq \sup |f| \cdot \epsilon$$

Since (*) holds for any refinement of P .

Exercise 6

#1. Suppose α increase on $[a, b]$, and continuous at $x_0 \in [a, b]$.

If $f(x_0) = 1$ and $f(x) = 0$ for all $x \neq x_0$, then $f \in \mathcal{B}(\alpha)$ and $\int f d\alpha = 0$.

Proof. Let $P_n := \{a, x_0 - \frac{1}{n}, x_0, x_0 + \frac{1}{n}, b\} \cap [a, b]$.

Given $\epsilon > 0$, there exists $\delta > 0$ s.t. $|x_0 - t| < \delta \Rightarrow |\alpha(x_0) - \alpha(t)| < \frac{\epsilon}{2}$

Now, for $N \in \mathbb{N}$ s.t. $1 < N\delta$, now,

$$0 = L(f, P_N, \alpha) \leq \int f d\alpha \leq \bar{\int} f d\alpha \leq U(f, P_N, \alpha) < \epsilon.$$

#2. Suppose that for some $x_0 \in [a, b]$, $f(x_0) > 0$.

Then, there exists $\delta > 0$ s.t. $|t - x_0| < \delta \Rightarrow |f(t) - f(x_0)| < \frac{1}{2} \cdot f(x_0)$.

By the triangle inequality $-\frac{1}{2}f(x_0) + |f(x_0)| < f(t) < \frac{1}{2}f(x_0) + |f(x_0)|$, if $|t - x_0| < \delta$.

Let $P \in \mathcal{P}[a, b]$ be any partition. Then, for a refinement

$$P^* = P \cup \{x_0 - \delta, x_0 + \delta\} \cap [a, b],$$

$$\text{Now, } \frac{f(x_0)}{2} \cdot \delta \leq U(f, P^*) \leq U(f, P)$$

But, we can take $P' \in \mathcal{P}[a, b]$ s.t. $U(f, P') < \frac{f(x_0)}{2} \cdot \delta$ since $\int_a^b f = 0$, it is contradiction.

#3. Define $\beta_1, \beta_2, \beta_3$ on $[-1, 1]$ s.t.

$$\beta_3(x) = 0 \text{ if } x < 0, \quad \beta_3(x) = 1 \text{ if } x > 0, \text{ and } \beta_1(0) = 0, \beta_2(0) = 1, \beta_3(0) = \frac{1}{2}.$$

Let f be a bounded map on $[-1, 1]$.

a). $f \in \mathcal{R}(\beta_1)$ if and only if $f(0^+) = f(0)$ and thus then $\int f d\beta_1 = f(0)$

b).

c). $f \in \mathcal{R}(\beta_3)$ if and only if f is continuous at 0.

d). If f is continuous at $x=0$, then

$$\int f d\beta_1 = \int f d\beta_2 = \int f d\beta_3 = 0.$$

Proof. Let $P_n = \{-1, 0, \frac{1}{n}, 1\}$. Then, for each $i=1, 2, 3$,

$$U(f, P_n, \beta_i) = M_1 \cdot [\beta_i(0) - \beta_i(-1)] + M_2 \cdot [\beta_i(\frac{1}{n}) - \beta_i(0)] + M_3 \cdot [\beta_i(1) - \beta_i(\frac{1}{n})]$$

$$= M_1 \beta_i(0) + M_2 [1 - \beta_i(0)] + 0$$

$$= (M_1 - M_2) \beta_i(0) + M_2 \quad \text{Similarly}$$

$$L(f, P_n, \beta_i) = (m_1 - m_2) \beta_i(0) + m_2.$$

When $i=1$, $\beta_1(0)=0$, therefore

$$U(f, P_n, \beta_1) - L(f, P_n, \beta_1) = M_2 - m_2 = \sup_{t \in [0, \frac{1}{n}]} f(t) - \inf_{t \in [\frac{1}{n}, 1]} f(t) \rightarrow 0 \text{ as } n \rightarrow \infty,$$

since $f(0^+) = f(0)$. To show b), take $P_n = \{-1, -\frac{1}{n}, 0, \frac{1}{n}, 1\}$. Then,

$$U(f, P_n, \beta_i) = M_1 [\beta_i(-\frac{1}{n}) - \beta_i(-1)] + M_2 [\beta_i(0) - \beta_i(-\frac{1}{n})] + M_3 [\beta_i(\frac{1}{n}) - \beta_i(0)] + M_4 [\beta_i(1) - \beta_i(\frac{1}{n})]$$

$$= M_2 \beta_i(0) + M_3 (1 - \beta_i(0)). \quad \text{Similarly}$$

$$L(f, P_n, \beta_i) = m_2 \beta_i(0) + m_3 (1 - \beta_i(0))$$

When $i=2$, $\beta_2(0)=1$. Therefore

$$U(f, P_n, \beta_2) - L(f, P_n, \beta_2) = M_2 - m_2 \rightarrow 0 \text{ if } f(0^-) = f(0).$$

When $i=3$, $\beta_3(0)=\frac{1}{2}$, then

$$U(f, P_n, \beta_3) - L(f, P_n, \beta_3) = \frac{1}{2} [M_2 - m_2 + M_3 - m_3] \rightarrow 0 \text{ if } f(0^+) = f(0) = f(0^-).$$

Moreover, $U(f, P_n, \beta_i) \rightarrow f(0) \cdot [\beta_i(0) + (1 - \beta_i(0))] = f(0) \text{ as } n \rightarrow \infty. \quad \square$

